

in the economy are affected—it is just a question of degree. Most consumers dislike low growth and high inflation because it is more likely they will be laid off or they are less able to afford the same basket of goods and services in real terms. A few investors, such as debt collectors, benefit from slow growth, and a few other investors, including owners of oil wells, benefit from high inflation induced by surging commodity prices. In general, though, bad outcomes of macro factors define bad times for the average investor.

The *level* of the factor often does not matter as much as a *shock* to a factor. Many macro factors are persistent: when inflation is low today, we know that it will be very likely low next month. The fact that it is then low at the end of the month is no surprise. What is surprising are any movements in inflation not anticipated at the beginning of the period. Thus, we often need to look at *unexpected changes* to macro factors.

Asset prices respond to these factors *contemporaneously*. As inflation is increasing or unexpected adverse inflation shocks hit the economy, we enter a bad time and asset prices fall. The risk premium over the long run compensates investors for the losses endured when bad times of high inflation occur in the short run.

The three most important macro factors are growth, inflation, and volatility.

2.1. ECONOMIC GROWTH

Risky assets generally perform poorly and are much more volatile during periods of low economic growth. However, government bonds tend to do well during these times. If an investor is in a position to weather recessions relatively comfortably, then that person should tilt more heavily toward risky assets such as equities. In doing so she'll enjoy higher returns, on average, and over the long run these will make up for losses during periods of low growth.⁴ If an investor cannot bear large losses during recessions, she should hold more bonds, especially government bonds. Her portfolio will likely perform much better during recessions but worse over the long run. This is the price the investor pays for low exposure to growth risk.

Table 7.2 reports means and volatilities of large stocks, small stocks, government bonds, and corporate bonds (investment grade and high yield) conditional on economic recessions and expansions defined by the National Bureau of Economic Research (NBER). I also report means and volatilities conditional on low and high real GDP growth and low and high consumption. These are computed simply by dividing the sample into two sets, above and below the median, respectively. Table 7.2 shows that, during recessions, stock returns fall: the mean return for large stocks is 5.6% during recessions and 12.4% during expansions. The difference in returns across recessions and expansions is more pronounced for the

⁴ A related variable to GDP growth is real consumption growth. It turns out that real consumption is very smooth and actually does not vary much across recessions and expansions, unlike GDP growth. We will look at the effects of consumption risk in chapter 8 when we discuss the equity risk premium.